Turbulent Mixing and NO_x Formation in Gas Turbine Combustors

ALEXANDER VRANOS
Pratt and Whitney Aircraft, 400 Main Street, East Hartford, Connecticut

An order of magnitude analysis is presented which shows that large-scale mixing in gas turbine combustors controls the stoichiometry of the combustion process at typical engine operating conditions. Fuel lean combustion, a desirable condition for low NO_x-formation rates, is thereby precluded for those combustors where fuel and air are fed separately. This is due to the fact that combustion times in the flammable range are very much shorter than mixing time. Low-pressure combustor test data indicate a fixed rate of NO_x generation consistent with a mixing rate-limited mechanism. Significant levels of free carbon in lean mixtures support the conclusion.

Introduction

In a conventional gas turbine combustor liquid fuel is injected directly into the combustion zone where the fuel is simultaneously dispersed, vaporized, and burned with air. In order to attain maximum performance, the equivalence ratio in the primary zone is approximately stoichiometric. An undesirable consequence of this type of operation is that high local temperatures are ensured, and the NO_x-formation rate is relatively high. Chemical kinetics predict an exponential reduction in NO_x formation rate with equivalence ratio for lean mixtures. However, it has been demonstrated [1, 2] that, in conventional gas turbine combustors, the reduction in NO_x level achieved by simply lowering primary zone equivalence ratio is small compared to the predictions of chemical kinetics. Friswell [1], for example, varied primary zone equivalence ratio over a four fold range by varying fuel flow while maintaining air flow constant and found that the relationship between NO_x level and equivalence ratio was simply linear. Certainly, this is a clear indication that mixing plays an important role in the NO_x-formation process. Pompei and Heywood [2], working with a simulated gas turbine combustor, found a strong dependence of NO_x concentration on equivalence ratio when mixing was accelerated by injecting high velocity air into the combustion zone. However, the relationship between NO_x and equivalence ratio deviated considerably from predictions of chemical kinetics. Heywood and coworkers [2, 3] also present some interesting theoretical calculations which relate NO_x levels to the mixedness of the combustion zone.

It is the purpose of this paper to discuss further the limitations imposed by mixing, vaporization, and combustion on NO_x formation and to present some new chemical composition data obtained from the combustion zone of a gas turbine combustor. It is demonstrated that appreciable lean combustion is not realized under normal operating conditions.

Discussion

In a conventional gas turbine combustor, NO_x is formed at a rate which is determined by the following processes: (1) mixing of fuel, air, and combustion products, (2) fuel vaporization (if applicable), (3) combustion, and (4) nitrogen fixation. These processes can be viewed as occurring sequentially or simultaneously depending upon the length of time associated with each process. There will be a "characteristic" time associated with each

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process, and when characteristic times are the same order of magnitude the processes overlap. On the other hand, when time scales differ by an order of magnitude the chance of overlap is small, and the process with the largest characteristic time controls. Four time scales have been considered for the NO_x-formation process, and each has been estimated for typical aircraft gas turbine operating conditions ($P = 220$ psia, $T_{inlet} = 800°F$, JP5, fuel). The time scales are estimated as follows:

(1) τ_m, the turbulent mixing time—a measure of the time required to dissipate concentration fluctuations and produce mixing on a molecular scale. Corrsin [4] shows that in a homogeneous, isotropic, stationary, turbulence field, the magnitude of a concentration fluctuation decreases exponentially with time, and τ_m is, by definition, the time constant for this exponential approach to uniformity. Corrsin derives the following expression between mixing time, τ_m, kinematic viscosity ν, and the Taylor microscale, λ:

$$\tau_m = \frac{\lambda^2}{\sigma\nu} \tag{1}$$

Corrsin also derives an expression for τ_m based on the properties of a turbulent mixer, where mixing time, τ_m (as defined above), mass of fluid in the mixer, M, characteristic dimension of the mixer, L, and power input to the mixer, P, are related by the expression:

$$\tau_m \alpha \left(\frac{ML^2}{P}\right)^{1/3} \tag{2}$$

In this paper, Eq. (2) has been used to estimate τ_m in a stoichiometric combustion zone. In order to use Eq. (2), typical combustor dimensions and operating conditions have been used to estimate M, L, and P. A proportionality constant has been obtained from data on stirred tank reactors. The power input, P, has been calculated by assuming that the kinetic energy feed rate of the combustion jets furnishes the power for stirring the combustion zone, and a typical combustor liner-pressure drop of 2.25% was assumed in order to compute this energy feed rate. The characteristic dimension of the combustor primary zone, L, has been assumed equal to 0.5 ft. M, the mass of fluid in the mixer, has been estimated from gas density and typical primary zone volume. Using Van De Vusse's [5] data for stirred tank reactors, Evangelista et al. [6] deduced that the value of the proportionality constant is approximately 10.0. The calculated value of τ_m is 10 m-sec.

(2) τ_v, vaporization time—the time for burning droplets to vaporize completely. τ_v has been estimated for various droplet sizes by solving, simultaneously, the momentum and vaporization equations for a single droplet burning in a constant velocity air stream of uniform properties. Spalding's [7] single droplet vaporization equation, corrected for forced convection by the Ranz and Marshall correlation [8], was used to compute vaporization rate. Droplet dynamics were computed using the Reynolds number-drag coefficient relationship of Dickerson et al. [9]. In accordance with data obtained from burning sprays and monosized streams [10], the single droplet vaporization times computed in this way have been multiplied by a factor of two in order to complete the estimate of vaporization time in a burning spray.

(3) τ_c, combustion time—the time required for a well mixed, gaseous, fuel-air mixture to release its chemical energy by molecular scale processes. τ_c has been estimated by dividing laminar combustion wave thickness by laminar burning velocity. Pertinent data have been taken from Ref. [11].

(4) τ_{NO}, nitrogen fixation time—τ_{NO} is defined by the relationship,

$$(\tau_{NO})^{-1} = \frac{1}{[NO]_{equi}} \frac{d[NO]}{dt},$$

where $d[NO]/dt$, the rate of formation of NO, is evaluated at the initial condition $(NO) \equiv 0$. The rate of formation has been computed from the kinetics model of Heywood et al. [12]. Heywood's model assumes that the concentrations $[N_2]$, $[O]$, $[O_2]$, $[OH]$, and $[H]$ are equilibrium values and the concentrations $[N]$ and $[N_2O]$ are steady/state values. Therefore, the effect of superequilibrium amounts of O atom on NO-formation rate is neglected. It is felt, however,

at this quasiequilibrium model is satisfactory at gh pressure because of the wide separation in ne scales between the nitrogen reactions and the mbustion reactions (13). Assuming, momenrily, that when air, gaseous fuel, and combustion oducts are brought together a distribution of xture ratios can be formed, characteristic times ve been estimated at typical engine operating nditions for a range of mixture ratios. The reits are shown in Table 1.

It is seen that the characteristic combustion nes are much shorter than the characteristic xing time. This confirms that the estimated xing time, calculated independently in this per, is of the correct order of magnitude since erall reaction times in gas turbine combustors e known to be about two orders of magnitude ger than would be predicted by "stirred reactor" netics. Furthermore, the disparity in time scales ecludes the formation of a wide range of mixres as originally postulated. To see this, it is nvenient to view the mixing process as one in hich air continually dilutes the fuel. At mixture tios above the rich flammability limits $\tau_m \ll \tau_c$ y definition), and dilution proceeds without mbustion. Near the rich flammability limit $\rightarrow \tau_c$, and dilution and combustion proceed nultaneously. However, further dilution causes to decrease rapidly relative to τ_m (as shown in ble 1), and combustion proceeds rapidly relative dilution. Therefore, air is burned rapidly as it utes the fuel, and unburned mixtures below the h flammability limits are not formed. Appreble NO formation is thereby precluded on the h side of stoichiometric. Appreciable NO rmation can occur on the lean side only for ose burned mixtures for which $\tau_{NO} < \tau_m$; from Table 1 this is true for $1.0 > \phi > 0.95$. Outside this range, on the fuel lean side, further dilution after consumption of the fuel "freezes" NO chemistry. The combination of rapid hydrocarbon and nitrogen chemistry within narrow concentration limits will force the NO production to proceed at a fixed rate, or equivalently, the total amount of NO formed should be proportional to the amount of fuel burned. This expectation is borne out by experimental data presented in later paragraphs.

TABLE 1

	τ_m	τ_c	τ_{NO}
7	10 m-sec	0.20 m-sec	330 m-sec
3	10	0.17	53
9	10	0.15	15
0	10	0.13	7
1	10	0.16	23
2	10	0.2	109
4	10	0.4	>1000

With regard to liquid fuels, the vaporization process impedes the molecular mixing process, and the arguments for mixing controlled NO_x generation are fortified. A comparison of times for vaporization, mixing and combustion are shown in Table 2.

TABLE 2

Drop Diameter	Liquid Fuels τ_v	τ_m	τ_c[a]
150 μ	9.8 m-sec	10 m-sec	0.13 m-sec
100	3.6	10	0.13
50	1.2	10	0.13
25	0.4	10	0.13
10	0.2	10	0.13

[a]Combustion time is assumed to be that of a stoichiometric mixture.

It is seen that, for $d < 100\ \mu$, $\tau_v < \tau_m$ and vaporization is completed prior to mixing so that diffusive gaseous burning still obtains. For $d < 10\ \mu$, $\tau_v \sim \tau_c$ and premixed combustion might occur were it not for, again, slow mixing.

Chemical Composition Measurements in a Gas Turbine Combustor

In order to confirm the ideas presented above, chemical composition measurements were made in the reaction zone of a gas turbine combustor. The test conditions were: $P = 1.3$ atm, $T_A = 785°F$, and $F/A = 0.0193$. The fuel was JP-5. The combustor was annular with six fuel nozzles disposed about the center line. A schematic cross-section of the combustor at a fuel nozzle location is shown in Fig. 1. Figure 1 also shows the probing stations and calculated air flows as percentages of

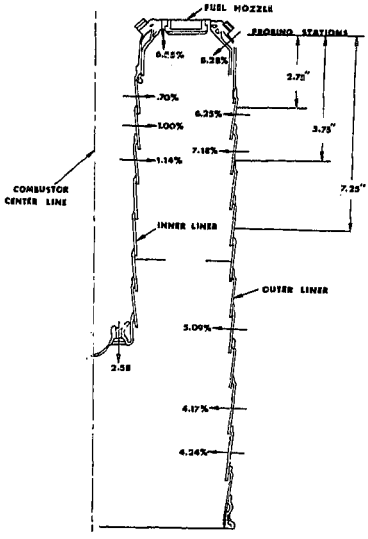

Fig. 1. Combustor schematic of cross-section.

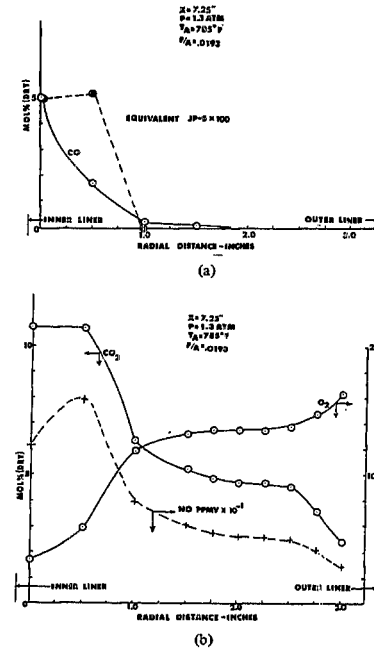

Fig. 2. Composition profiles. $X = 7.25''$, $P = 1.3$ atm, $T_A = 785°F$, $F/A = 0.0193$.

the total air flow (excluding cooling airflows). Traverses were made at each axial location from the outer liner to the inner liner using a steam-cooled probe in line with the fuel injector. Air holes were also in line with the fuel injector. The following species were measured: oxygen, hydrogen, carbon monoxide, carbon dioxide, nitric oxide, nitrogen dioxide, and hydrocarbons (measured as total equivalent methane). Despite the rapid rate of air addition and the small characteristic dimension of the passage, the data indicate a high level of unmixedness. Typical composition profiles are given in Fig. 2. These profiles indicate that there are steep concentration gradients across the passage, and the fuel is displaced toward the centerbody by the flow of the air (see CO_2, CO, and JP-5 profiles). The composition profiles are similar to those obtained from diffusion flames, and it follows that there are large temperature gradients and high temperature zones in the passage.

All of the comments above apply to the macroscopic behavior of the system. Inferences may be made about the microscopic behavior of the system by examining Fig. 3, the composition correlation. Figure 3 is a plot of NO_x concentration vs CO_2 concentration for all radial and axial positions. The CO_2 concentration is taken as a measure of the amount of burned fuel in the mixture. A linear relationship obtains below 9% CO_2, independent of sampling location. (Because of the large amount of scatter in the upper portion of the curve, it is not obvious that a strictly linear relationship holds over the entire range of the data. Above 9% CO_2, the carbon monoxide concen-

TABLE 3
Calculated Carbon Concentrations $X = 7.75''$

Moles carbon per 100 miles gaseous mixture	0.797	0.923	1.425	1.51	1.2	2.22	2.66
Local equivalence ratio	0.28	0.274	0.292	0.296	0.276	0.276	0.24

tration rises rapidly, and the CO_2 concentration alone is not a good indicator of the burnedness of the mixture.) Since the straight line begins at the origin, the NO_x/CO_2 ratio is nearly constant over the entire range of the data. This behavior is not attributed to entrainment of uniformly mixed gases from the recirculation zone, because it is thought that most of the fuel is consumed external to the recirculation zone. Rather, the observed behavior is consistent with the mixing rate controlled mechanism presented in previous paragraphs. The constancy of the NO_x/CO_2 ratio is

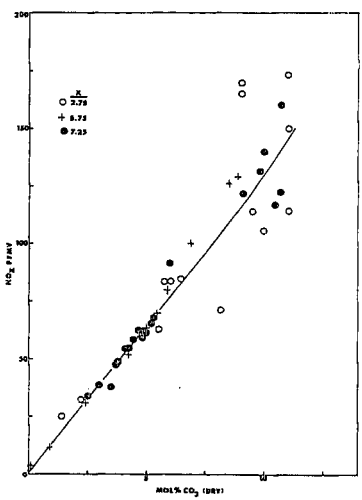

Fig. 3. Composition correlation. ○ $X/2.75''$, + 5.75, ● 7.25".

indicative of a process wherein the following conditions are both fulfilled: (1) combustion occurs over a very narrow range of equivalence ratios and, (2) subsequent dilution of the combustion products proceeds, on a molecular level, at a rate which is invariant within the reaction volume. When sufficient dilution of the combustion products has occurred, the nitrogen chemistry is frozen, and the NO_x/CO_2 ratio remains constant as the mixture is diluted further.

Another interesting aspect of the chemical composition measurements is that the amount of carbon present can be computed. Some results are presented in Table 3. Carbon has been computed assuming that there are no partially oxidized species except carbon monoxide and all the hydrogen is either free, in the form of water, or in unreacted fuel. The most interesting aspects of this table are: (1) the large amount of free carbon (1-2%) and (2) the existence of carbon in areas which are fuel lean, even below the lower flammability limit. Taken together, these facts imply a high level of unmixedness in the system, since carbon is not formed in lean premixed mixtures.

Conclusion

From the foregoing analysis it is concluded that for combustors in which fuel and oxidant are fed separately, the NO_x formation is dependent primarily on the amount of fuel burned and not on the overall mixture ratio. This is because combustion times in the flammable range are very much shorter than mixing times, making the reaction rates dependent primarily on mixing rate. This is borne out by the data from a gas turbine annular combustor. Carbon formation in very lean overall mixtures supports the conclusion.

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